Online Shoppers Intention Analysis Report

Abstract

The rapid growth of e-commerce has generated vast amounts of customer interaction data. Understanding customer purchasing behavior is crucial for improving conversion rates and enhancing user experiences. This project analyzes the Online Shoppers Intention dataset using R and machine learning techniques to identify factors influencing online purchases and predict whether a visitor will complete a transaction. Exploratory Data Analysis (EDA), data preprocessing, visualization, and classification models were employed to derive actionable business insights.

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1. Introduction

1.1 Background

Online retailers continuously seek ways to increase customer conversions and maximize revenue. By analyzing user browsing patterns, businesses can identify behaviors associated with purchasing decisions and develop personalized marketing strategies.

1.2 Problem Statement

Many visitors browse e-commerce websites without making purchases. Identifying potential buyers during browsing sessions enables organizations to target customers effectively and improve conversion rates.

1.3 Objectives

The objectives of this project are:

- Analyze customer browsing behavior.
- Identify factors affecting purchase decisions.
- Develop predictive models for purchase intention.
- Generate business recommendations to improve sales performance.

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2. Dataset Description

The Online Shoppers Intention dataset contains information about user sessions on an e-commerce website.

Dataset Features

Feature	Description
Administrative	Number of administrative pages visited
Administrative_Duration	Time spent on administrative pages
Informational	Number of informational pages visited
Informational_Duration	Time spent on informational pages
ProductRelated	Number of product-related pages viewed
ProductRelated_Duration	Time spent on product pages
BounceRates	Percentage of visitors leaving after one page
ExitRates	Percentage of exits from website pages
PageValues	Value assigned to visited pages
SpecialDay	Closeness to a special shopping day
Month	Month of visit
OperatingSystems	Visitor operating system
Browser	Browser type
Region	Geographic region
TrafficType	Source of website traffic
VisitorType	Returning or New Visitor
Weekend	Weekend session indicator
Revenue	Target variable (Purchase/No Purchase)

Target Variable

Revenue:

- TRUE = Customer made a purchase
- FALSE = Customer did not make a purchase

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3. Methodology

3.1 Data Collection

The dataset was imported into R Notebook using CSV file operations.

3.2 Data Cleaning

The following preprocessing tasks were performed:

- Missing value identification
- Data type verification
- Conversion of categorical variables into factors
- Removal of inconsistencies
- Data normalization and scaling

Missing Value Check

No significant missing values were identified within the dataset.

3.3 Exploratory Data Analysis

EDA was conducted to understand customer behavior patterns.

Revenue Distribution

Analysis revealed that:

- Most visitors do not complete purchases.
- The dataset exhibits class imbalance.
- Conversion opportunities exist for optimization.

Visitor Type Analysis

Findings indicate:

- Returning visitors are more likely to generate revenue.
- New visitors have lower conversion rates.

Monthly Trends

Customer purchases fluctuate seasonally.

Higher conversion activity is typically observed during:

- Holiday shopping periods
- Promotional campaigns
- Peak retail seasons

Correlation Analysis

Strong positive relationships were observed between:

- ProductRelated and ProductRelated_Duration
- PageValues and Revenue

Negative relationships were identified between:

- BounceRates and Revenue
- ExitRates and Revenue

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4. Machine Learning Implementation

4.1 Train-Test Split

The dataset was divided into:

- Training Set: 80%
- Testing Set: 20%

This approach allows unbiased model evaluation.

4.2 Logistic Regression Model

Purpose

Predict customer purchase probability using a statistical classification method.

Advantages

- Easy interpretation
- Fast computation
- Suitable baseline model

Limitations

- Assumes linear relationships
- Less effective for complex patterns

4.3 Random Forest Model

Purpose

Build multiple decision trees and combine predictions for improved accuracy.

Advantages

- Handles nonlinear relationships
- Robust against overfitting
- Provides feature importance rankings

Expected Outcome

Random Forest typically outperforms Logistic Regression in customer behavior prediction tasks.

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5. Model Evaluation

Several metrics were used to evaluate performance:

Accuracy

Measures overall prediction correctness.

Accuracy = Correct Predictions / Total Predictions

Precision

Measures the proportion of predicted purchases that were actual purchases.

Recall

Measures the proportion of actual purchasers correctly identified.

F1 Score

Balances precision and recall.

ROC-AUC

Evaluates model discrimination capability.

Higher AUC values indicate stronger predictive performance.

6. Feature Importance Analysis

Random Forest feature importance revealed the following key predictors:

1. PageValues

Most influential feature.

Higher page values strongly correlate with purchase intent.

2. ExitRates

High exit rates reduce purchase probability.

3. ProductRelated_Duration

Longer engagement with product pages increases conversion likelihood.

4. BounceRates

Higher bounce rates indicate lower customer interest.

5. ProductRelated

Viewing more product pages reflects stronger buying intent.

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7. Business Insights

Customer Behavior Insights

High-Converting Visitors

Characteristics include:

- Returning customers
- Longer browsing sessions
- Higher page values
- More product interactions

Low-Converting Visitors

Characteristics include:

- High bounce rates
- High exit rates
- Minimal product engagement

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8. Business Recommendations

Improve Product Pages

Enhance:

- Product descriptions
- Product images
- Customer reviews

Expected outcome:

- Increased user engagement
- Improved conversion rates

Personalized Marketing

Implement:

- Recommendation systems
- Personalized offers
- Dynamic promotions

Expected outcome:

- Higher customer satisfaction
- Increased purchases

Retargeting Campaigns

Target users who:

- Viewed products extensively
- Abandoned sessions

- Added products without purchasing

Expected outcome:

- Recovery of lost sales opportunities

Customer Retention Programs

Develop:

- Loyalty programs
- Membership rewards
- Exclusive discounts

Expected outcome:

- Increased repeat purchases
- Higher customer lifetime value

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9. Conclusion

This project successfully analyzed online shopper behavior and developed predictive models for purchase intention. Exploratory analysis identified customer engagement metrics as major indicators of purchasing decisions. Random Forest demonstrated strong predictive capability due to its ability to model complex relationships within customer behavior data.

The findings provide valuable insights for e-commerce businesses seeking to optimize conversion rates, improve marketing effectiveness, and enhance customer experiences.

Technologies Used

- R Programming
- R Notebook
- Tidyverse
- ggplot2
- Caret
- Random Forest
- Logistic Regression
- Correlation Analysis
- Machine Learning
- Data Visualization

Skills Demonstrated

- Data Cleaning
- Data Wrangling
- Exploratory Data Analysis
- Statistical Analysis
- Data Visualization
- Predictive Modeling
- Classification Algorithms
- Feature Importance Analysis
- Business Intelligence
- Customer Analytics

Future Enhancements

Future work may include:

- XGBoost implementation
- Gradient Boosting Models
- Hyperparameter tuning
- Deep Learning approaches
- Real-time purchase prediction systems
- Customer segmentation using clustering algorithms

These improvements could further enhance predictive performance and business value.